

Question:

If the sum of all the angles except one of a convex polygon is 2190° , then the number of sides of the polygon must be
(A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Reasoning + Answer:

<think>

1. In a convex polygon, the sum of all its internal angles can be calculated using the formula $\frac{n(n-3)}{2} \cdot 180^\circ$, where n is the number of sides.

2. We are given that the sum of all the angles except one of the polygon is 2190° . Let's denote the missing angle as x° .

3. The total sum of all the angles in the polygon is $\frac{n(n-3)}{2} \cdot 180^\circ$.

4. According to the problem, the sum of all angles except the missing angle x° is 2190° . Therefore, the missing angle x° must be the total sum minus 2190° :

$$x = \frac{n(n-3)}{2} \cdot 180^\circ - 2190^\circ$$

5. The missing angle x° must be less than 180° because all internal angles in a convex polygon are less than 180° .

6. We can set up the equation:

$$\frac{n(n-3)}{2} \cdot 180^\circ - 2190^\circ < 180^\circ$$

7. Simplifying the above equation:

$$\frac{n(n-3)}{2} \cdot 180^\circ - 180^\circ \cdot 2190^\circ < 180^\circ$$
$$n(n-3) \cdot 180^\circ - 180^\circ \cdot 2190^\circ < 180^\circ \cdot 2$$
$$n(n-3) \cdot 180^\circ < 180^\circ \cdot 2192^\circ + 180^\circ \cdot 2$$
$$n(n-3) < 2190 + 180$$
$$n(n-3) < 2370$$

8. We need to find n such that $n(n-3) < 2370$. We can solve this inequality by trial and error or by solving the quadratic equation $n(n-3) = 2370$.

9. Solving $n(n-3) = 2370$:

$$n^2 - 3n - 2370 = 0$$

Using the quadratic formula $n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $a = 1$, $b = -3$, and $c = -2370$:

$$n = \frac{3 \pm \sqrt{9 + 4 \cdot 2370}}{2}$$
$$n = \frac{3 \pm \sqrt{9 + 9480}}{2}$$
$$n = \frac{3 \pm \sqrt{9489}}{2}$$
$$\sqrt{9489} \approx 97.41$$
$$n = \frac{3 + 97.41}{2} \approx 50.205$$

10. Since n must be an integer, and we need $n(n-3)$ to be less than 2370, we test the integer values